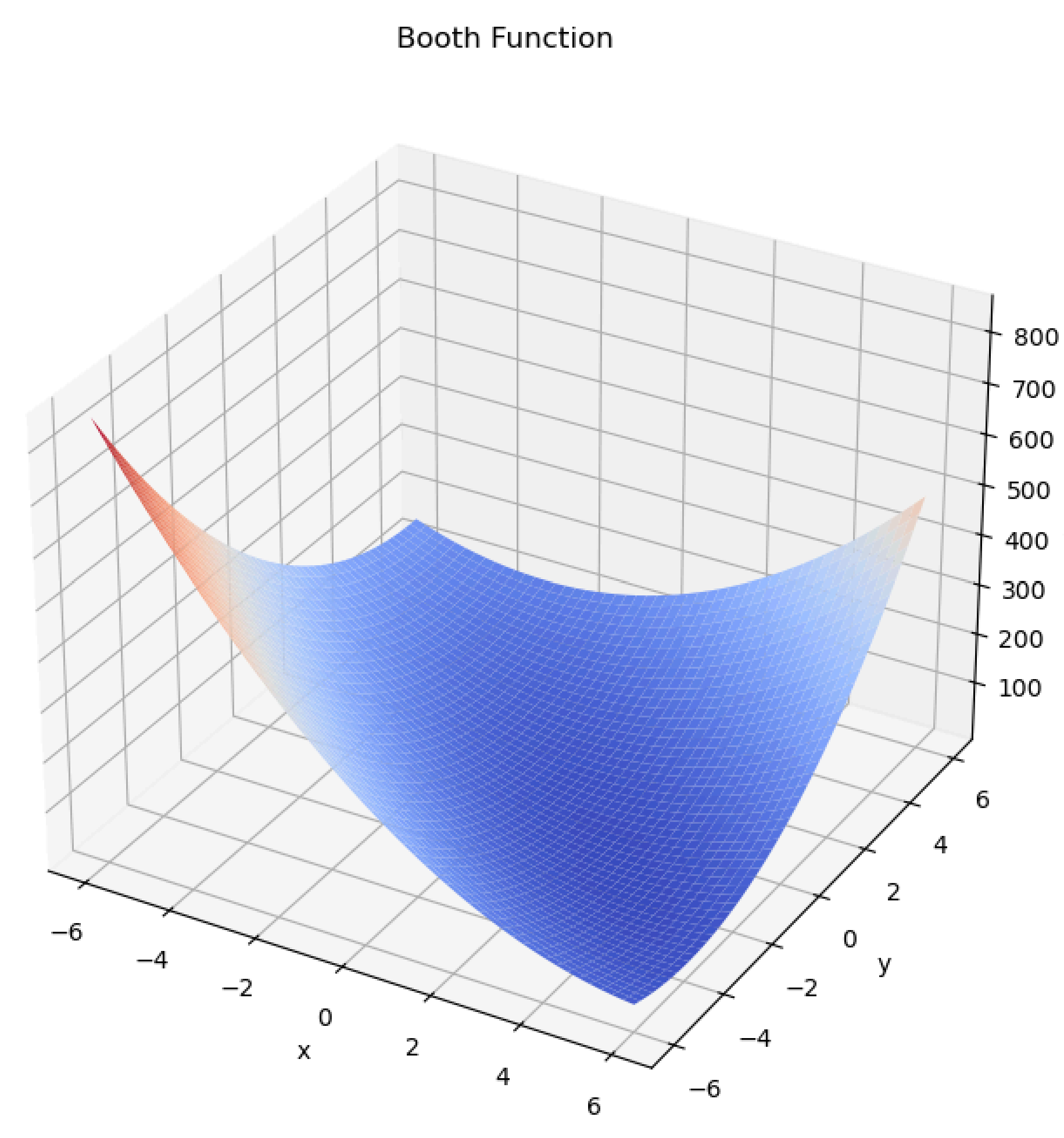
In [24]:
```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits import mplot3d
from matplotlib import cm
```

In [50]:
```python
# cleaning a booth fuction method
def booth_function(x,y):
    return (x + 2*y)**2 + (2*x + y - 5)**2
```

In [26]:
```python
# Data for a three-dimensional line
x = np.linspace(-6, 6, 50)
y = np.linspace(-6, 6, 50)
xline, yline = np.meshgrid(x,y)
z = booth_function(xline, yline)

fig = plt.figure(figsize =[12, 8])
ax = plt.axes(projection='3d')
ax.plot_surface(xline,yline,z, cmap = cm.coolwarm)
ax.set_xlabel("x")
ax.set_ylabel("y")
ax.set_zlabel("f(x, y)")
ax.set_title('Booth Function')
```

Out[26]:
```
Text(0.5, 0.92, 'Booth Function')
```



In [27]:
```python
# Derivative of booth functions
def partial_derivative_x(x, y):
    return 2 * (x + 2 * y) + 4 * (2 * x + y - 5)

def partial_derivative_y(x, y):
    return 4 * (x + 2 * y) + 2 * (2 * x + y - 5)
```

In [28]:
```python
# Gradient Descent initialization
alpha = 0.01  # Learning rate
initial_x, initial_y = -1, 1  # Initial guess
tolerance = 1e-6  # Tolerance for stopping condition
max_iters = 1000  # Maximum number of iterations

x, y = initial_x, initial_y
path_x, path_y = [x], [y]  # Store the path for plotting

#loop for gradient descent
for i in range(max_iters):
    # Compute the gradient
    dfdx = partial_derivative_x(x, y)
    dfdy = partial_derivative_y(x, y)

    # Update x and y
    x_new = x - alpha * dfdx
    y_new = y - alpha * dfdy

    # Check for convergence
    if abs(x_new - x) < tolerance and abs(y_new - y) < tolerance:
        print(f"Converged after {i} iterations.")
        break

    # Update variables for next iteration
    x, y = x_new, y_new
    #appending each x and y for path so that it is plotted well
    path_x.append(x)
    path_y.append(y)
```

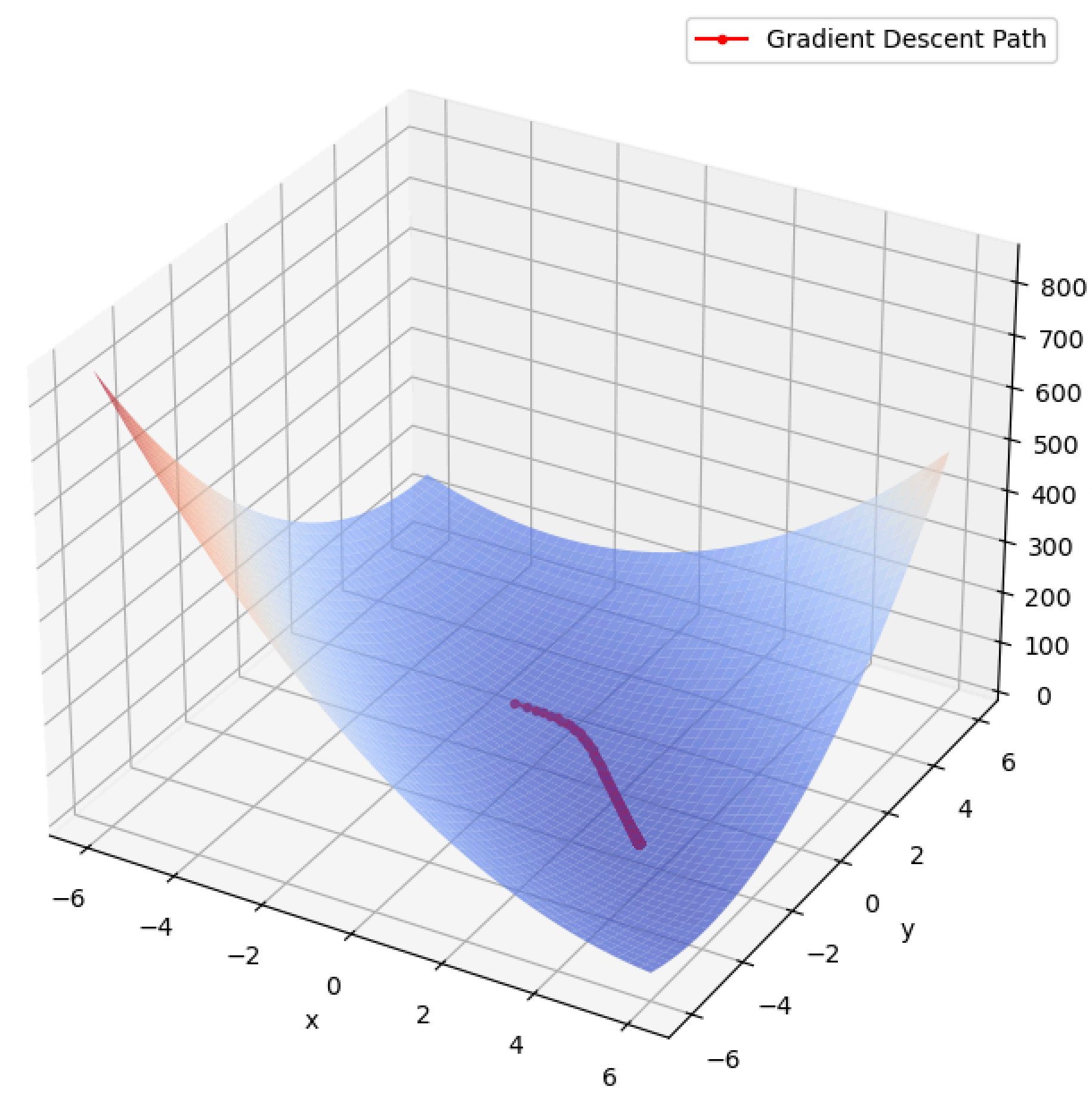
```
Converged after 553 iterations.
```

In [29]:
```python
fig = plt.figure(figsize =[12, 8])
ax = plt.axes(projection='3d')
ax.plot_surface(xline,yline,z,cmap = cm.coolwarm, alpha=0.7)
ax.set_xlabel("x")
ax.set_ylabel("y")
ax.set_zlabel("f(x, y)")

# Calculate the function values along the path for plotting
path_z = booth_function(np.array(path_x), np.array(path_y))

# Plot the gradient descent path
ax.plot(path_x, path_y, path_z, color='r', marker='o', markersize=3, label="Gradient Descent Path")
ax.legend()

plt.show()
```



In [ ]:

In [56]:
```python
def backtracking_line_search(x, y, dfdx, dfdy, alpha=1.0, beta=0.5, gamma=0.5):
    while True:
        # Check the Armijo condition (sufficient decrease condition)
        new_x = x - alpha * dfdx
        new_y = y - alpha * dfdy
        if booth_function(new_x, new_y) <= booth_function(x, y) - gamma * alpha * (dfdx**2 + dfdy**2):
            break  # Accept the step size
        alpha *= beta  # Reduce the step size
    return alpha

# Initialize starting point
x, y = initial_x, initial_y

# Gradient Descent Loop with Backtracking Line Search
for i in range(max_iters):
    # Compute the gradient at the current point
    dfdx = partial_derivative_x(x, y)
    dfdy = partial_derivative_y(x, y)

    # Determine adaptive step size using backtracking line search
    alpha = backtracking_line_search(x, y, dfdx, dfdy)

    # Update x and y based on the gradient and adaptive learning rate
    x_new = x - alpha * dfdx
    y_new = y - alpha * dfdy

    # Check for convergence: stop if updates are smaller than the tolerance
    if abs(x_new - x) < tolerance and abs(y_new - y) < tolerance:
        print(f"Converged after {i+1} iterations.")
        break

    # Update current values of x and y for the next iteration
    x, y = x_new, y_new

# If the loop completes without convergence
else:
    print("Gradient descent did not converge within the maximum number of iterations.")
```

```
Converged after 37 iterations.
```

In [ ]:

In [31]:
```python
#loading the data and viewing the first 5 rows
data = pd.read_csv('regression.csv')
data.head(5)
```

Out[31]:

	X1	X2	X3	X4	X5	X6	X7	X8	X9	X10	X11	Y
0	7.4	0.70	0.00	1.9	0.076	11.0	34.0	0.9978	3.51	0.56	9.4	5
1	7.8	0.88	0.00	2.6	0.098	25.0	67.0	0.9968	3.20	0.68	9.8	5
2	7.8	0.76	0.04	2.3	0.092	15.0	54.0	0.9970	3.26	0.65	9.8	5
3	11.2	0.28	0.56	1.9	0.075	17.0	60.0	0.9980	3.16	0.58	9.8	6
4	7.4	0.70	0.00	1.9	0.076	11.0	34.0	0.9978	3.51	0.56	9.4	5

In [ ]:

In [6]:
```python
#checking for any missing values
data.isnull().sum()
```

Out[6]:
```
X1     0
X2     0
X3     0
X4     0
X5     0
X6     0
X7     0
X8     0
X9     0
X10    0
X11    0
Y      0
dtype: int64
```

In [7]:
```python
data.describe()
```

Out[7]:

	X1	X2	X3	X4	X5	X6	X7	X8	X9	X10	X11	Y
count	1599.000000	1599.000000	1599.000000	1599.000000	1599.000000	1599.000000	1599.000000	1599.000000	1599.000000	1599.000000	1599.000000	1599.000000
mean	8.319637	0.527821	0.270976	2.538806	0.087467	15.874922	46.467792	0.996747	3.311113	0.658149	10.422983	5.636023
std	1.741096	0.179060	0.194801	1.409928	0.047065	10.460157	32.895324	0.001887	0.154386	0.169507	1.065668	0.807569
min	4.600000	0.120000	0.000000	0.900000	0.012000	1.000000	6.000000	0.990070	2.740000	0.330000	8.400000	3.000000
25%	7.100000	0.390000	0.090000	1.900000	0.070000	7.000000	22.000000	0.995600	3.210000	0.550000	9.500000	5.000000
50%	7.900000	0.520000	0.260000	2.200000	0.079000	14.000000	38.000000	0.996750	3.310000	0.620000	10.200000	6.000000
75%	9.200000	0.640000	0.420000	2.600000	0.090000	21.000000	62.000000	0.997835	3.400000	0.730000	11.100000	6.000000
max	15.900000	1.580000	1.000000	15.500000	0.611000	72.000000	289.000000	1.003690	4.010000	2.000000	14.900000	8.000000

In [8]:
```python
data.shape
```

Out[8]:
```
(1599, 12)
```

In [9]:
```python
#Train to Test split should be 80-20.
# Shuffle and split the DataFrame
train_data = data.sample(frac=0.8, random_state=42)  # 80% for training
test_data = data.drop(train_data.index)  # Remaining 20% for testing
```

In [18]:
```python
#normalization of data
def standard_scaler(x_data, X_column):
    mean = np.mean(X_column)
    std_dev = np.std(X_column)
    z = (x_data - mean) / std_dev
    return z
```

In [19]:
```python
#creating a new data empty datafarne to store the standardized data
train_data_standardized = pd.DataFrame()

for column in train_data.columns:
    train_data_standardized[column] = standard_scaler(train_data[column].values ,train_data[column] )
```

In [20]:
```python
test_data_standardized = pd.DataFrame()

for column in test_data.columns:
    test_data_standardized[column] = standard_scaler(test_data[column].values ,test_data[column] )
```

In [57]:
```python
import numpy as np

# Optimization class to implement different optimization methods
class Optimization:
    def __init__(self, X, y):
        self.X = X
        self.y = y
        self.loss = Loss(X, y)  # Instantiate the Loss class

    # Newton's Method Optimization
    def newtons_method(self, initial_w, max_iters=1000, tolerance=1e-6):

        w = initial_w

        for i in range(max_iters):
            # Compute the gradient and Hessian matrix for the loss function
            gradient = self.loss.cross_entropy_loss_gradient(w)  # Gradient of the loss
            hessian = self.hessian_matrix(w)  # Hessian matrix of the loss function

            # Update the weights using Newton's method
            w_new = w - np.linalg.inv(hessian).dot(gradient)

            # Check for convergence
            if np.linalg.norm(w_new - w) < tolerance:
                print(f"Converged after {i+1} iterations.")
                break

            # Update the weights for the next iteration
            w = w_new

        return w

    # Hessian matrix for the loss function (Second derivative)
    def hessian_matrix(self, w):

        predictions = self.loss.sigmoid(np.dot(self.X, w))
        H = np.dot(self.X.T, self.X * predictions * (1 - predictions)) / len(self.y)
        return H
```

In [58]:
```python
# Loss class to handle different loss functions and their gradients
class Loss:
    def __init__(self, X, y):
        self.X = X
        self.y = y

    # Mean Squared Loss (MSE)
    def mean_squared_loss(self, w):
        predictions = np.dot(self.X, w)
        loss = np.mean((predictions - self.y) ** 2)
        return loss

    # Gradient of Mean Squared Loss
    def mean_squared_loss_gradient(self, w):
        predictions = np.dot(self.X, w)
        gradient = 2 * np.dot(self.X.T, (predictions - self.y)) / len(self.y)
        return gradient

    # Cross-Entropy Loss (Logistic Loss)
    def cross_entropy_loss(self, w):

        predictions = self.sigmoid(np.dot(self.X, w))
        loss = -np.mean(self.y * np.log(predictions + 1e-8) + (1 - self.y) * np.log(1 - predictions + 1e-8))
        return loss

    # Gradient of Cross-Entropy Loss
    def cross_entropy_loss_gradient(self, w):
        predictions = self.sigmoid(np.dot(self.X, w))
        gradient = np.dot(self.X.T, (predictions - self.y)) / len(self.y)
        return gradient

    # Sigmoid function for logistic regression
    def sigmoid(self, z):
        return 1 / (1 + np.exp(-z))
```

In [64]:

In [63]:

In [ ]:

In [ ]:

In [ ]: